UrbanLens — Milestone 1 Report

Urban Market Intelligence & Affordability Analytics

> IMPORTANT
>
> Build status: ✅ **PASSING —** `tsc -b && vite build` exits 0 · 2,261 modules · 1.95s

Git Checkpoints

Commit	Message
`2bf8666`	`feat: milestone 1 — initial UrbanLens dashboard`
`70ccc9c`	`fix: remove .tsx extension from main.tsx import for strict TS build`

Commands to Run the Application

```
bash
# Navigate to the project root
cd "/Users/vanshbarnes/Documents/UrbanLens Project"

# Start the frontend (Vite dev server — http://localhost:5173)
npm --prefix client run dev

# Start the backend (Express API — http://localhost:3001)
npm --prefix server run dev

# Production build
npm --prefix client run build
```

> TIP
>
> The Vite dev server auto-proxies `/api/*` requests to `http://localhost:3001`, so no CORS configuration is needed during development.

Files Created (37 total)

Root

File	Purpose
`package.json`	Workspace scripts (`dev:client` , `dev:server` , `build:client`)
`.gitignore`	Node, Vite dist, Python, OS artifacts
`README.md`	Architecture overview + setup instructions

Server (`server/`)

File	Purpose
`server/package.json`	Express, CORS, TypeScript, ts-node, nodemon
`server/tsconfig.json`	CommonJS / Node TS config
`server/src/index.ts`	Express server on `:3001` with `/api/health`

Client (`client/`)

Config & Setup

File	Purpose
`client/vite.config.ts`	Vite 6 + `@tailwindcss/vite` plugin + `/api` proxy
`client/index.html`	HTML entry with UrbanLens metadata
`client/tsconfig.json`	Solution-style TS config
`client/tsconfig.app.json`	Strict frontend TS (no `.tsx` in imports)
`client/tsconfig.node.json`	Vite config TS settings
`client/src/vite-env.d.ts`	Vite client types + PNG image module declarations
`client/src/index.css`	`@import "tailwindcss"` (v4 syntax) + custom scrollbars
`client/src/main.tsx`	React StrictMode root mount
`client/src/App.tsx`	BrowserRouter + collapsible sidebar layout + 7 routes

Synthetic Data

File	Purpose
`client/src/data/syntheticData.ts`	60-month price history, 5 Mumbai sectors, forecast, affordability brackets, rental yields — all labeled `// SYNTHETIC DEMO DATA`

Layout Components

File	Purpose
`client/src/components/layout/Sidebar.tsx`	Responsive collapsible sidebar (256px ↔ 72px), mobile drawer, DEMO MODE badge
`client/src/components/layout/Header.tsx`	Sticky header with route title, breadcrumb, ⚠️ Synthetic Data badge

Reusable UI Components

File	Purpose
`client/src/components/ui/MetricCard.tsx`	KPI card with value, Δ%, trend icon, Recharts sparkline
`client/src/components/ui/DemoNotice.tsx`	Dismissible amber disclaimer banner
`client/src/components/ui/EmptyState.tsx`	Centered placeholder with icon + CTA
`client/src/components/ui/LoadingSpinner.tsx`	Accessible animated spinner

Charts

File	Purpose
`client/src/components/charts/PriceTrendChart.tsx`	Multi-line Recharts `LineChart` · ₹/sq.ft tooltip · COVID & RBI reference lines
`client/src/components/charts/MarketOverviewChart.tsx`	Recharts `BarChart` for sector price comparison
`client/src/components/charts/AffordabilityChart.tsx`	Recharts `RadarChart` for multi-dimension livability

Pages (7 Routes)

Route	File	Description
`/`	`Overview.tsx`	4 KPI cards, 24-month price chart, bar chart, neighborhood snapshot table with gentrification badges
`/market-trends`	`MarketTrends.tsx`	Sector filter, date toggle, seasonality heatmap grid, transaction volume chart, stats table
`/forecasting`	`Forecasting.tsx`	SARIMA forecast chart (solid historical + dashed forecast + CI shading), 6m/12m projection cards, anomaly log
`/neighborhood`	`NeighborhoodIntelligence.tsx`	Two-column layout: amenity bars + infrastructure badges left, interactive Leaflet map right
`/affordability`	`Affordability.tsx`	3-tab: EMI Calculator · Rent vs Buy · Affordability Index
`/data-sources`	`DataSources.tsx`	Synthetic data transparency, data dictionary, production integration roadmap
`/methodology`	`Methodology.tsx`	Pipeline stages, EMI/SARIMA/GVI formulas, limitations

Dependencies Installed

Client (130 packages)

Package	Version	Purpose
`react` + `react-dom`	18.x	UI framework
`vite`	6.4.5	Build tool
`@vitejs/plugin-react`	latest	React fast refresh
`tailwindcss` + `@tailwindcss/vite`	4.x	Utility CSS (v4 Vite plugin)
`recharts`	2.15.4	Charts (LineChart, BarChart, RadarChart, AreaChart)
`leaflet` + `react-leaflet` + `@types/leaflet`	latest	Interactive maps
`react-router-dom` + `@types/react-router-dom`	6.x	Client-side routing
`lucide-react`	latest	Icon library
`clsx`	latest	Conditional className utility

Server (128 packages)

Package	Purpose
`express`	HTTP server
`cors`	Cross-Origin support
`typescript` + `ts-node` + `nodemon`	TS dev runtime

Build Output

```
vite v6.4.5 building for production...
✓ 2261 modules transformed.
dist/index.html                   1.01 kB │ gzip:   0.55 kB
dist/assets/index-CnAslV-z.css   49.18 kB │ gzip:  13.22 kB
dist/assets/index-DrrqnkL9.js   886.19 kB │ gzip: 250.07 kB
✓ built in 1.95s
```

> NOTE
>
> The 886 kB JS bundle warning is expected — Recharts + Leaflet + Lucide are large libraries. Code-splitting with dynamic `import()` per page will be addressed in a future milestone.

Issues Fixed

Issue	Fix Applied
`TS5097` : import path ends with `.tsx` without `allowImportingTsExtensions`	Removed `.tsx` extension from `main.tsx` → `import App from './App'`
Sandbox has no network → npm install silently hung	Re-ran npm install with network access (bypass sandbox)
Recharts v2 deprecation warning	Non-breaking warning; recharts v3 migration will be addressed in chart milestone

Known Limitations (Milestone 1 Scope)

Limitation	Planned Resolution
All data is 100% synthetic	Milestone 2: real data ingestion (NHB, OSM, Census)
ML pipeline not yet built	Milestone 3: Python SARIMA / scikit-learn pipeline
JS bundle not code-split (886 kB)	Future: dynamic imports per route
Server has only `/api/health`	Milestone 2: data API endpoints
No authentication	Out of scope per project spec
Recharts v2 (deprecated)	Upgrade to v3 in chart milestone
MongoDB not set up	Only needed when persistence is justified (Milestone 2+)

What's Next — Milestone 2

- Python data ingestion pipeline (`server/python/`)
- Real or realistic open datasets (NHB RESIDEX indices, OSM amenity data)
- Express API routes that serve processed data to the frontend
- Replace synthetic `priceTimeSeries` with API-driven data
- MongoDB integration (if persistence is justified)